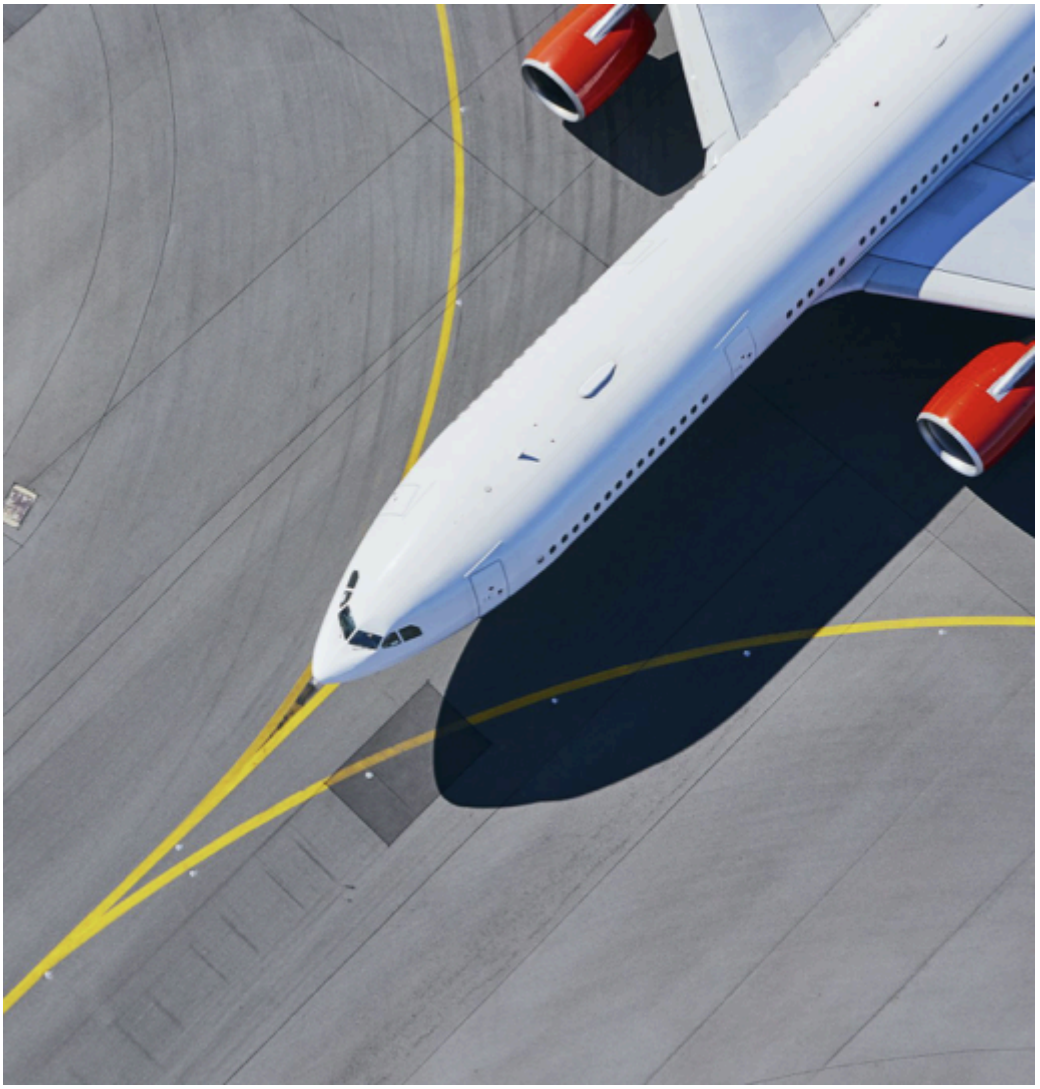

The Impact of Weather on Flight Delays

Capstone project

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Problem statement

Flight delays are not just an inconvenience — they represent a multi-billion-dollar operational challenge for the aviation industry. In the U.S. alone, delays have been estimated to cost around $33 billion yearly.

Whilst flight delays due to weather conditions is not something that could be fully prevented, there are benefits of predicting flight delays. Since aviation operations are chain operations and one aircraft often does multiple flights a day, being aware of an upcoming departure delay could give an airline a chance to prevent further network delays by aircraft substitution, engaging reserve crew, and repositioning staff. Other adjustments could include better fuel consumption planning and managing passenger expectations and therefore avoiding compensation claims.

Various studies have addressed the cost of delays issue. One of the studies, The University of Westminster (UoW) report, published in 2004 and updated in 2011 and 2015, is one of the most recent and comprehensive studies of the cost of delay in the air traffic management system in Europe. Explicit cost data, as extracted from University of Westminster study and adjusted to 2022 prices, indicates that a single 15 minute delay at the gate for an A321 aircraft could cost an airline €689 as a base tactical cost, disregarding the chain reaction effect. If we are able to create a model that would predict five 15 minute delays a week for this type of aircraft, the avoided tactical cost alone could be approximately €180,000 annually — excluding downstream network effects.

Industry/ domain

This project is situated within the commercial aviation and airport operations domain, where managing flight departure reliability is critical due to the high operational and customer costs associated with delays. The industry operates as a tightly coupled system involving airlines, airports, air traffic control, and ground services, meaning that disruptions such as any delays propagate quickly across the value chain. With increasing traffic volumes and greater weather variability, there is growing pressure to improve on-time performance through data-driven decision-making.

Stakeholders

Stakeholders include airlines, airports, air traffic management, and ground operations such as security, catering, and passenger services, all of whom are affected by departure delays either directly or through downstream network effects.

Business question

The objective of this project is to examine the extent to which weather conditions influence the likelihood of flight departure delays and to develop a predictive model using combined flight and meteorological data.

Such a model could support operational stakeholders across the aviation ecosystem enabling earlier planning and helping minimise the downstream impact of delays. This could include crew planning, aircraft routing, gate allocation, and passenger reaccommodation.

The goal is to catch delayed flights whilst also keeping an eye on the false positives as they also come at cost, so we will look at recall as well as average precision score for metrics.

Data question

The data question for the problem will be ‘ To what extent weather conditions influence the likelihood of flight departure delays?’ To answer this question, we will need data on flights with an indication of a delay and meteorological data and we will limit data to one airport of origin, and concentrate on the delays at gate only.

Since The United States Federal Aviation Administration (FAA) considers a flight to be delayed when it is 15 minutes later than its scheduled time, we will create a new column with 0 for no delay and 1 for delay when departure delay is over 15 mins. These delays may be influenced by weather as well as non-weather factors.

Data

The data was sourced from United States Department of Transportation, Bureau of Transportation Statistics, ‘On-Time: Reporting Carrier On-Time Performance (1987-present)’.

The website allows you to filter by geography (states and territories), year, and period, as well as select various fields with information on the flights, and you can then download the data in the csv format. For the purpose of this project we limited the geography to New York state. For our train data, we downloaded the 2022 and 2023 flight information. The test dataset included January to October 2025 data.

Given we were looking to examine the weather in the airport of origin, the fields we selected included:

- origin details,
- flight date
- scheduled departure time
- departure delay

We also included but did not use much the following fields:

- destination details
- cause of delay

Each csv file contained 17 columns. The 24 csv files for the train set were combined into one DataFrame, and the data was limited to JFK, which ended up being over 250,000 observations.

The Bureau of Transportation Statistics website currently contains data up to October 2025 but is constantly updated with new data. Some columns do include null values and require cleaning.

For the purpose of this project we disregarded how much time a plane might have sat on the runway due to a delay, we only examined scheduled departure time and the length of the delay.

The target variable is a column called ‘delayed_15’ which was created based on the ‘departure delay’ column, by assigning 1 for ‘delay’ for those observations where the departure delay is equal or over 15 mins. The percentage of the flights delayed for or over 15 mins in our train dataset is 24,4% and 20.3% for the test dataset.

Data science process

Data Cleaning

Once the data was limited to JFK, for both the train and test sets we performed the following:

- renamed ‘flight_date’ column
- cleaned up data formats since the columns in the DataFrame were a combination of objects, int64 and float64 types. ‘dep_time’ and ‘dep_delay’ columns were converted from float64 to int64 formats to allow for further data manipulation.
- identified the null values and approached them in various ways:
    - the cause of delay columns had lots of null values which were filled with 0
    - There were 7,528 observations missing values in the ‘dep_time’ and ‘dep_delay’ columns. Since the missing values did not allow us to see if those flights were delayed or not and there was no way to fill this missing information in, we dropped rows with these missing values.
- created a ‘scheduled departure time’ column that combined the flight date with scheduled departure time columns and the values in the new column are in the datetime64[ns] format.
- added meteorological data in using Meteostat Python library based on the scheduled departure time, hourly, for station_id = '74486' #John F. Kennedy International Airport (JFK). Most flights are matched within approx. 15 minutes of weather data, with the worst case being 30 minutes for the train set and 59 mins for the test set.

- dropped any rows with missing weather data. As a result, we ended up with 261,001 entries for train data and 85,520 for test data.
- The data then was checked again for any null values and any columns that had no data were dropped. For the train set it ended up being 'snwd', 'wpgt', 'tsun' since there were unfortunately no observations in these columns. The test dataset did include ‘wpgt’ observations, Peak Wind Gust data, but since the train and test data should have the same predictor columns for consistency of the model’s performance, the Peak Wind Gust data data was removed from the test model.
- In terms of any missing weather related values:
    - ‘prcp’ was missing 15537, fill with a 0
    - for the ‘cldc’, we created a new column indicating data was missing, then filled the missing data with the median of ‘cldc’
    - for ‘coco’, fill any missing data with 0 - it’s not a category as such, since ‘coco’ is a categorical value from 1 to 27.
- Once the data was cleaned, we created our target variable column, ‘delayed_15’, based on the ‘departure delay’ column, by assigning 1 for ‘delay’ for those observations where the departure delay is equal or over 15 mins.
- When preparing the data for modelling, we encoded categorical ‘coco’ variables to dummy variables and performed sine and cosine transformations for wind direction data

Our final DataFrames contain 261,001 observations in the train set, with 33 columns, and 85,435 observations in the test set, with 33 columns.

Data analysis

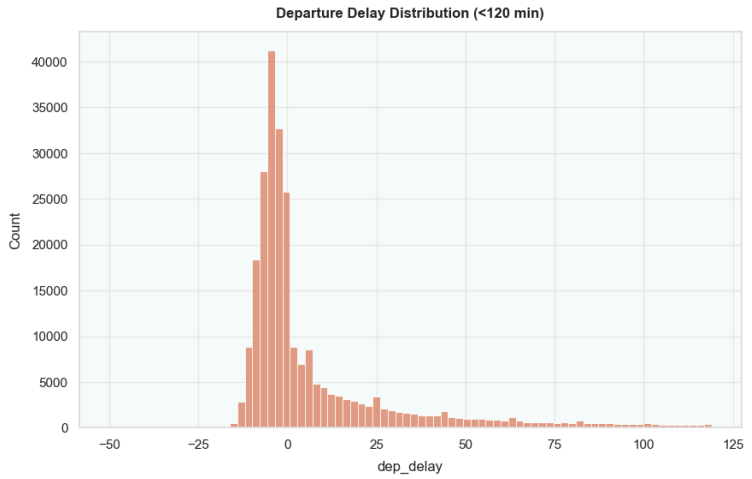
To perform the Exploratory Data Analysis (EDA), we identified which features were continuous features, ordinal features, circular features and categorical features and explored them accordingly.

Train dataset

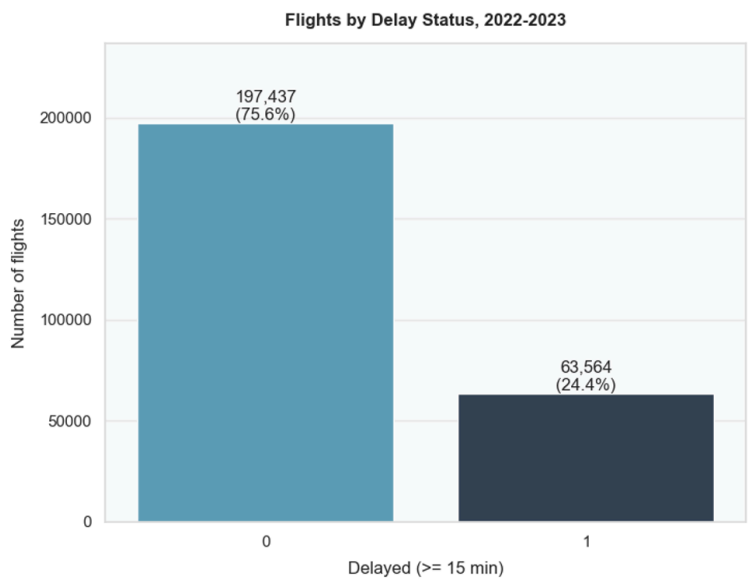
The statistical summary performed on the continuous features indicate that various features differ in range greatly, e.g. Total Precipitation ranges from 0 to 31.2, with the median being 0, whilst Air Pressure (MSL) ranges from 982.9 to 1041.4. Depending on the model, we need to scale our data.

	count	mean	std	min	25%	50%	75%	max
temp	261001.0	13.4028	9.124129	-15.0	6.1	13.3	21.1	35.0
rhum	261001.0	67.015019	20.191253	11.0	51.0	67.0	84.0	100.0
prcp	261001.0	0.111344	0.800417	0.0	0.0	0.0	0.0	31.2
wspd	261001.0	18.145338	9.472275	0.0	11.2	16.6	24.1	64.8
pres	261001.0	1016.918879	8.04015	982.9	1011.9	1016.8	1022.2	1041.4
dep_delay	261001.0	16.535147	56.927069	-50.0	-5.0	-2.0	14.0	1627.0

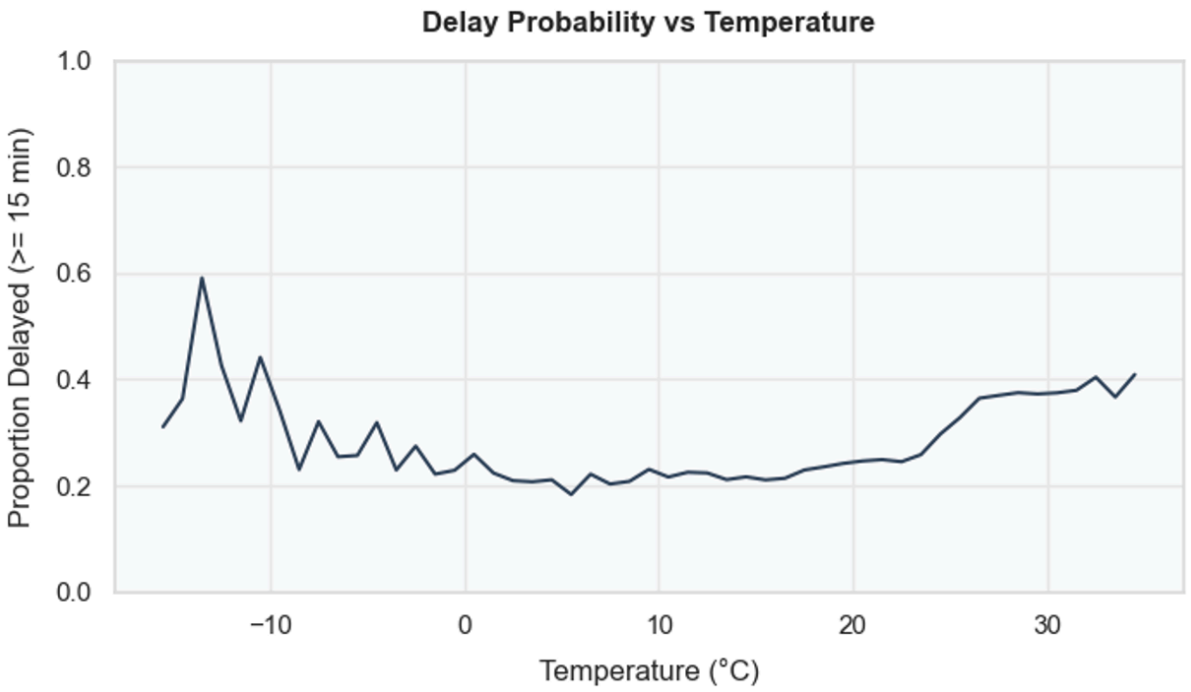
Included above is the 'dep_delay' column that we used to create our target variable. The statistical summary indicates that 50% of the flights actually depart the gate two minutes earlier than the scheduled departure time. 75% of the flights leave within 14 minutes of the scheduled departure time, but extreme tail exists.



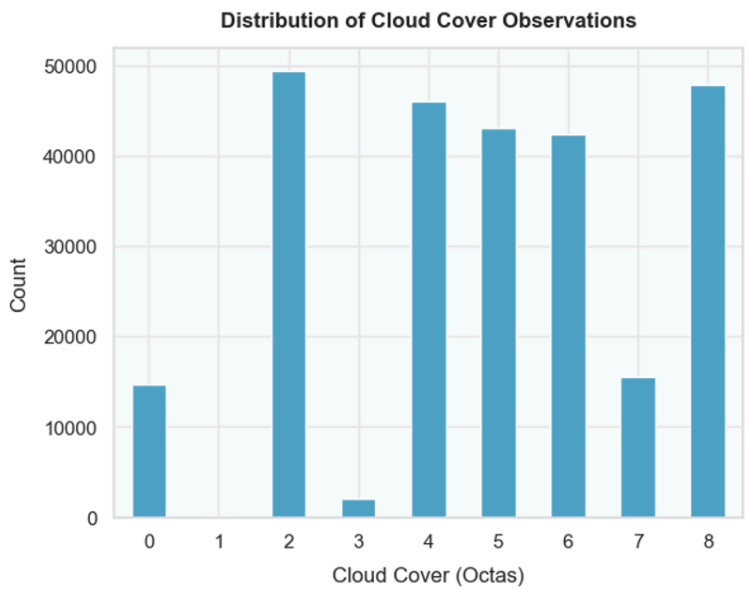
Based on the target variable we created, the proportion of delayed flights in the train dataset is 24.4%



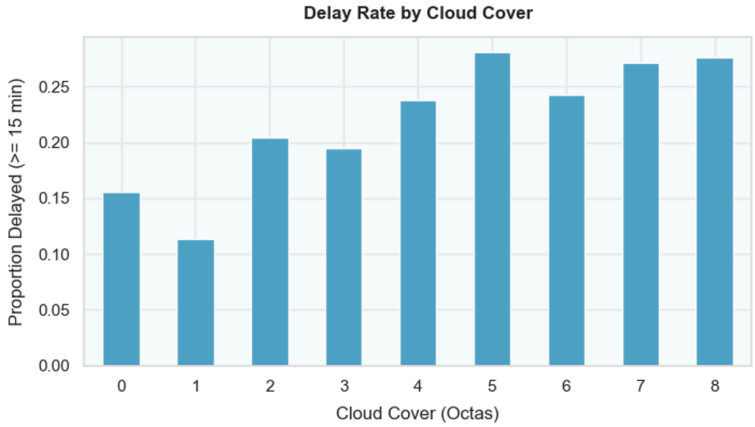
The temperatures range from -15 to 35 degrees Celsius. Our data indicates that there could be an increased probability of flights being delayed as the temperature drops or rises. This could be associated with any de-icing procedures in cold weather or for hot weather the fact that hot air reduces the lift of an aircraft.



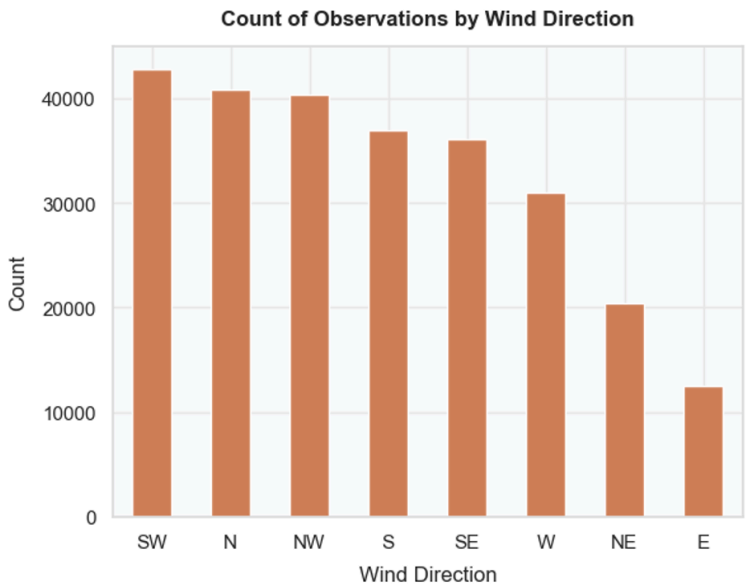
Cloud cover observations, they are ordinal, 0 is no cloud cover and 8 oktas represents full cloud cover with no breaks. The dataset includes a variety of observations.



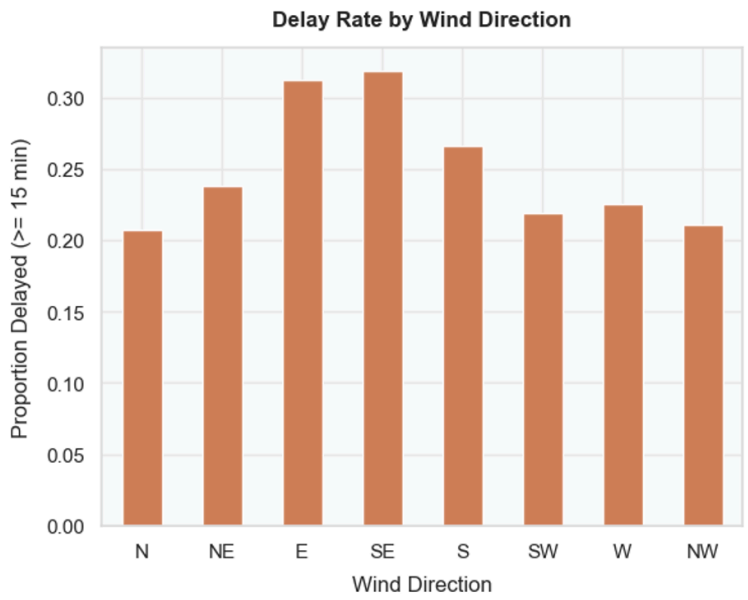
The train data indicates that flights departing under high cloud cover (7–8 octas) experience an approximately 10–12 percentage-point higher probability of departure delay (>15 minutes) compared to clear-sky conditions.



As to the wind direction, there is a range of observations in the training set.

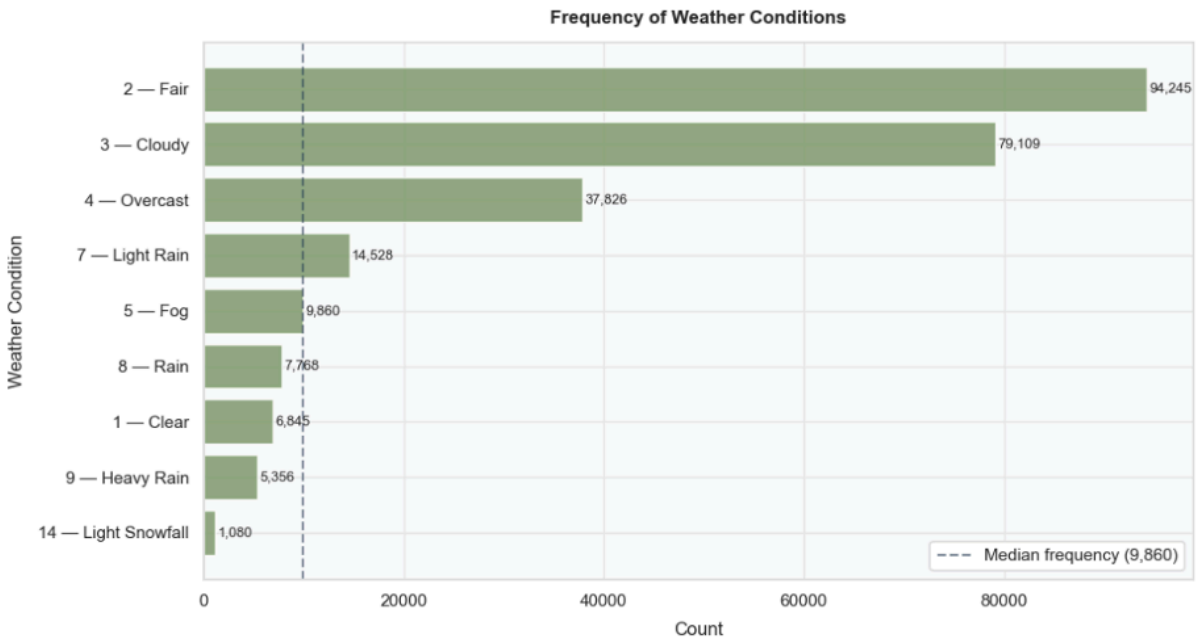


The above graph shows that the probability of departure delay increases with the eastern and south-eastern winds. For our location, JFK, Eastern and north-eastern winds are often associated with coastal systems, storms, nor'easters.

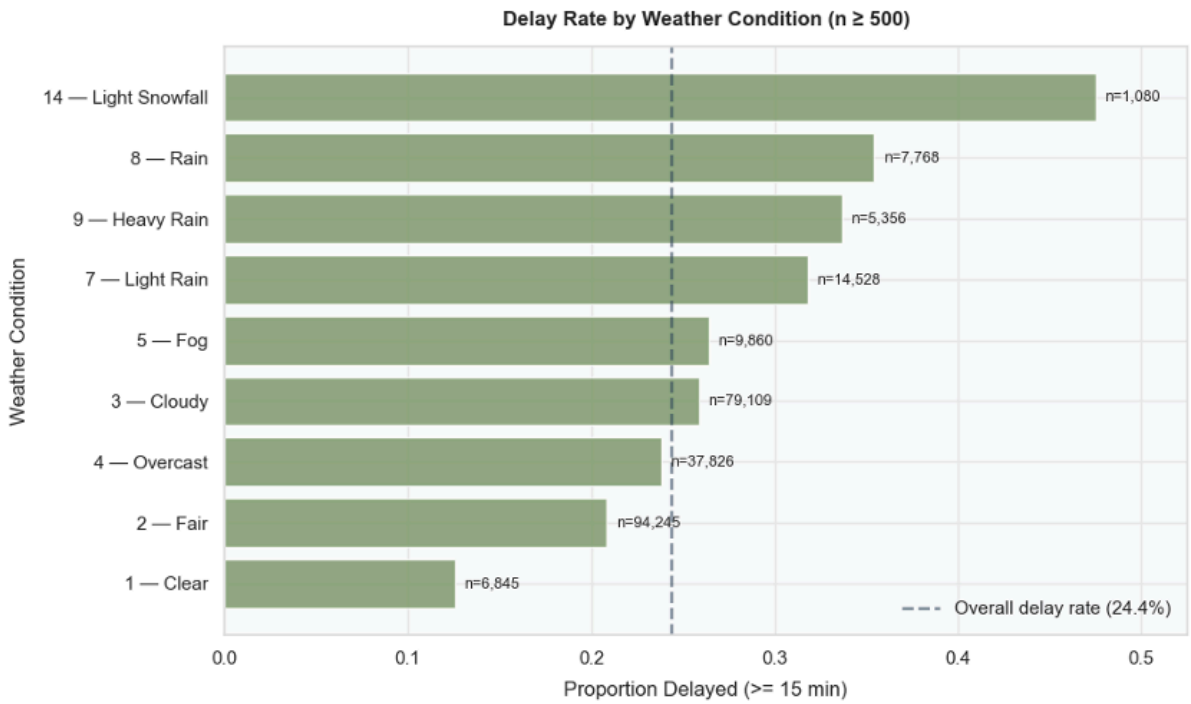


Given JFK's runway configurations and regional meteorology, wind direction likely reflects broader weather patterns and operational conditions rather than acting as an isolated causal factor.

With the weather conditions code, 99% of observations range from Clear to Heavy Rain.



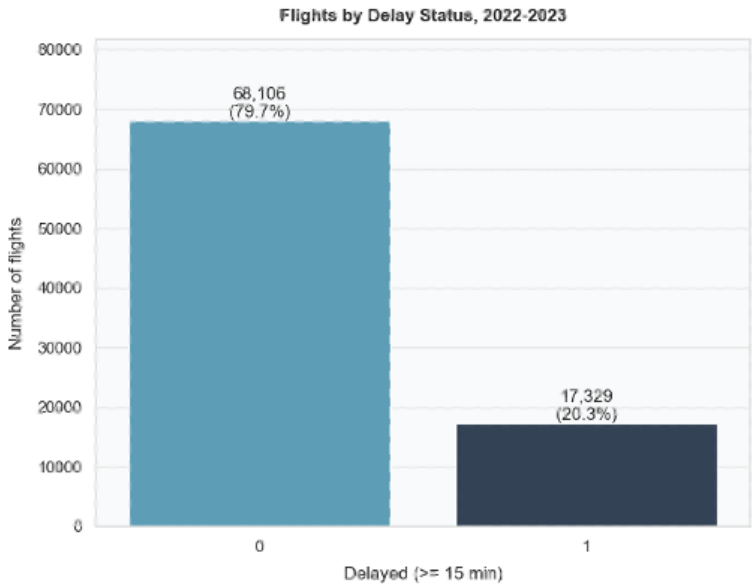
Within the remaining 1% is 62% of the observations are Light Snowfall and we can see that flights departing under this weather code condition experience an approximately 12-13 percentage-point higher probability of departure delay (>15 minutes) compared to other weather conditions.



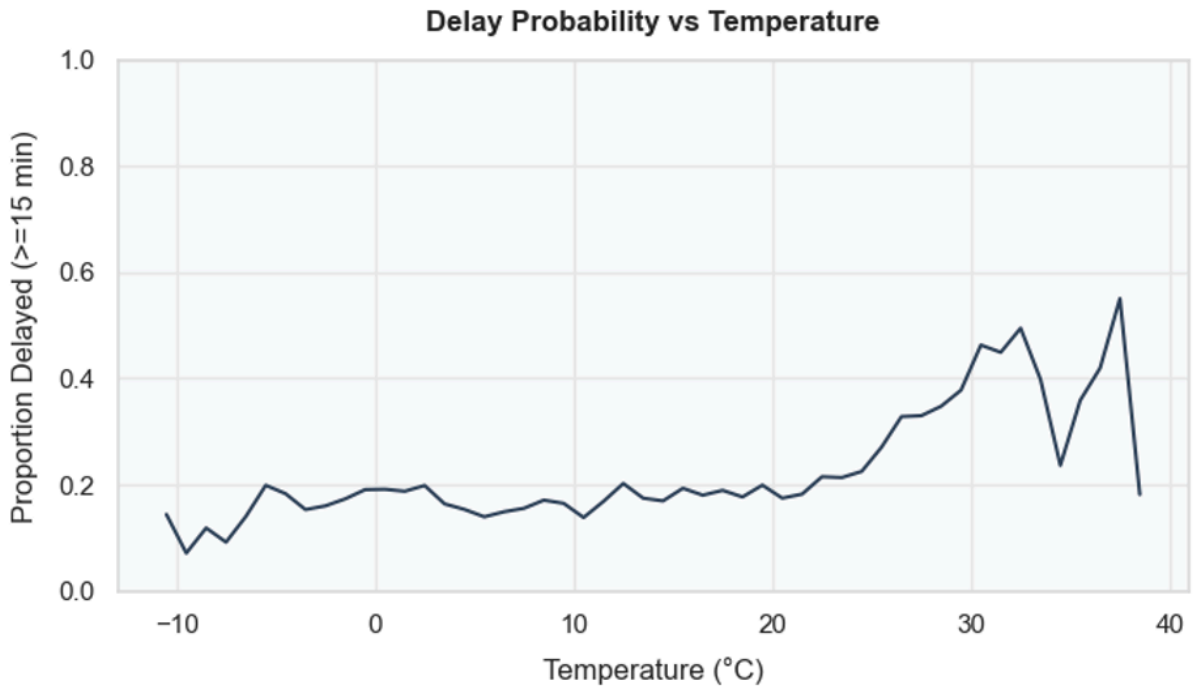
The absence of other Weather Condition Codes is most likely linked to the fact that severe weather conditions don't just cause delays but cancellations.

Test dataset

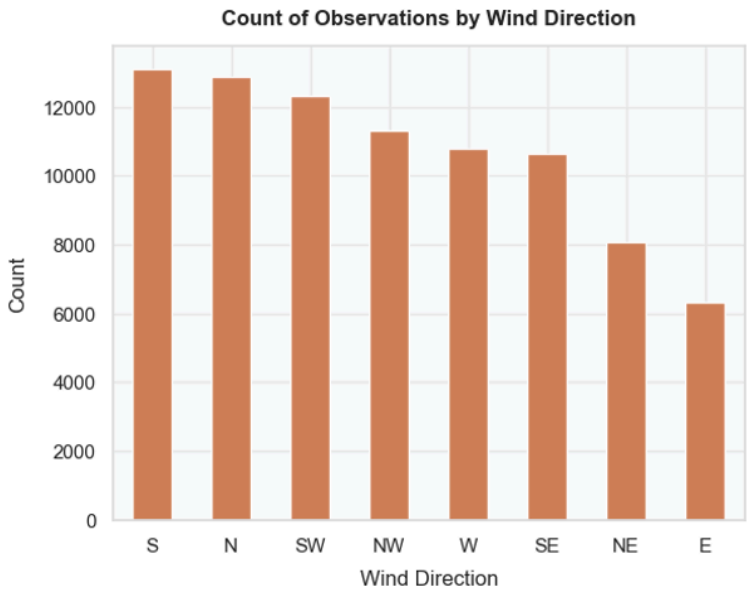
For the test data, the January to October 2025 flights, the proportion of delayed flights is 20.3%.



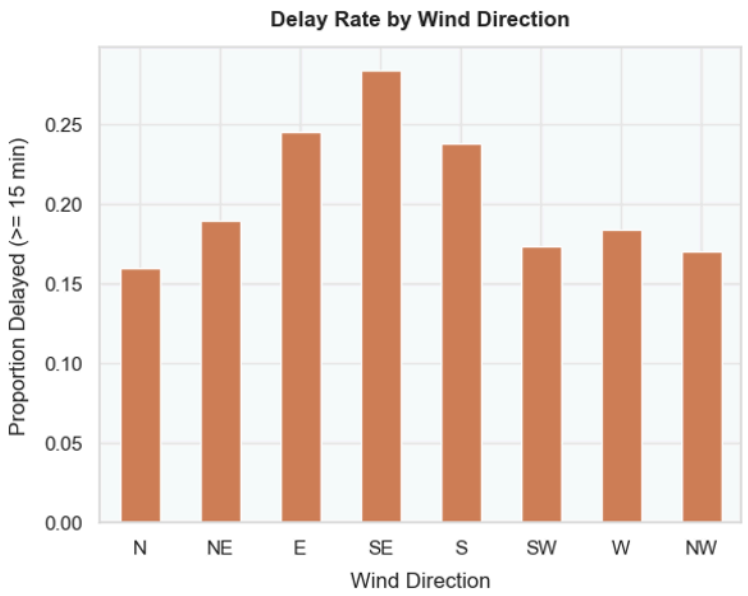
Temperatures range from -10 to 38.9 degrees Celsius. That is a true observation, the Eastern US experienced a rare June heatwave event in 2025. On June 24 and 25, 2025, JFK Airport recorded 102°F, marking the hottest June day on record for that location.



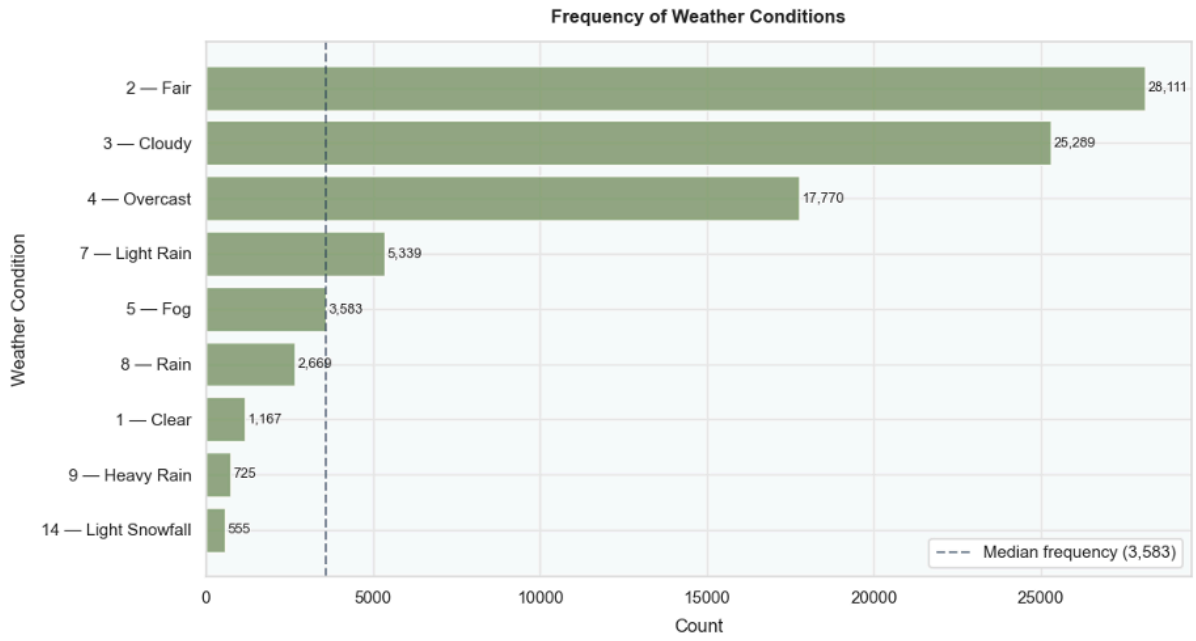
Slightly different distribution of the wind directions.

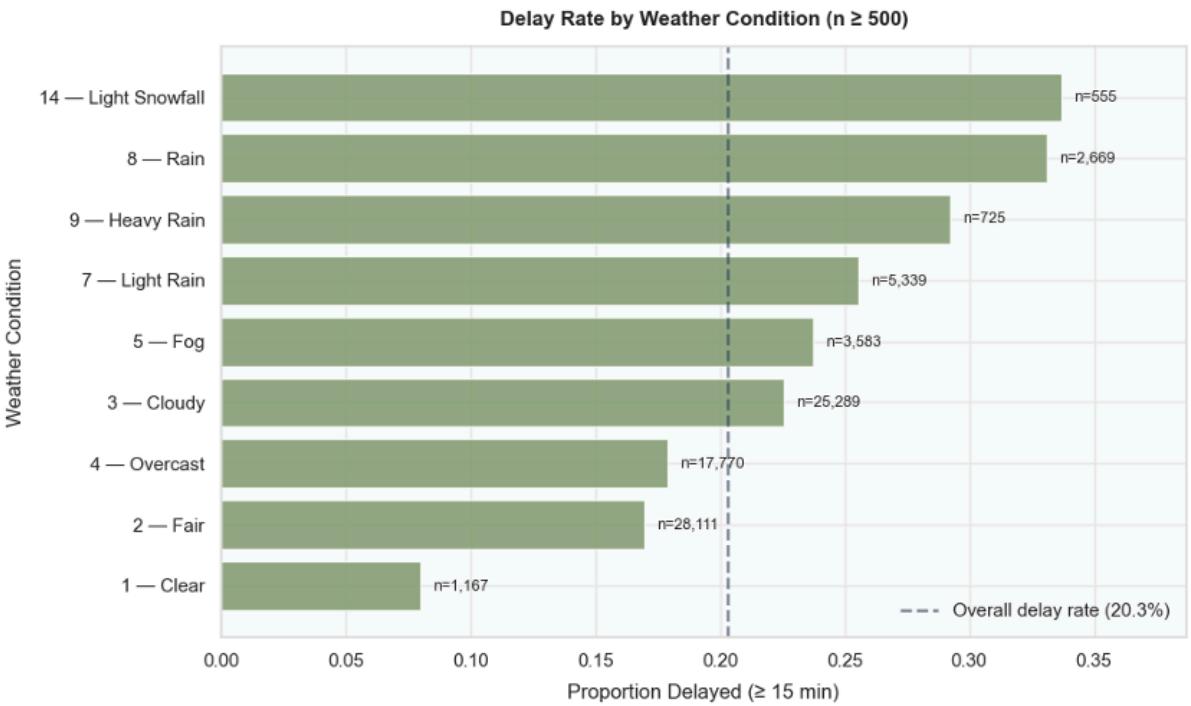


However, we see again the increased probability of delays with eastern and south-eastern winds.



With regards to the wind direction, 99% of observations again range from Clear to Heavy Rain. The remaining 1% is made up of 70% of the observations are Light Snowfall.





We again see the probability of flight delays increasing with the light snowfall.

Overall, weather condition distributions were highly consistent between train and test sets, suggesting a stable atmospheric regime and supporting the temporal robustness of the model.

Modelling

Features and target

In terms of the features and target, the predictor columns include:

- Continuous features such as air temperature, relative humidity, total precipitation, wind speed, air pressure
- Ordinal features such as cloud cover
- Circular features such as wind direction (sine and cosine transformations)
- A categorical feature called weather condition code (one-hot encoded)

The target variable is ‘delayed_15’.

Models used

We have selected the following models for this project: Logistic Regression, Decision Tree, Random Forest, XGBoost. No cloud platforms were used for this project.

All of the models were trained on 2022-2023 data and tested them on 2025 data. For Logistic Regression, the data was scaled.

Logistic Regression, Decision Tree, and Random Forest were all optimised with GridSearchCV. Random Forest GridSearchCV takes approximately 25-30 mins to run, and other models were much faster.

Results evaluation

Our objective was to identify as many delayed flights as possible whilst controlling false positives, which also carry operational cost.

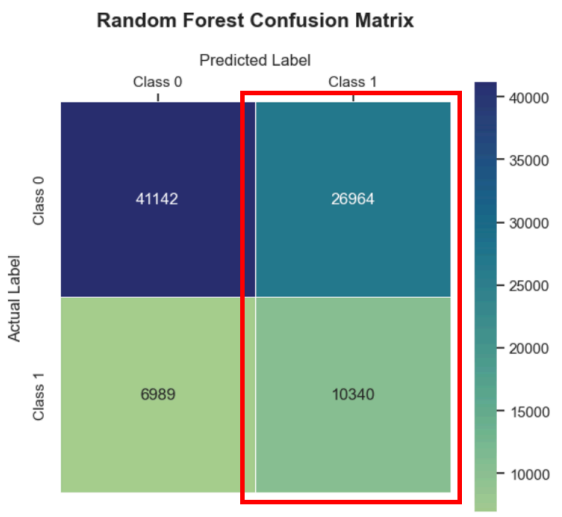
Therefore, model evaluation prioritised recall for delayed departures alongside Precision–Recall AUC to assess overall performance on the minority class.

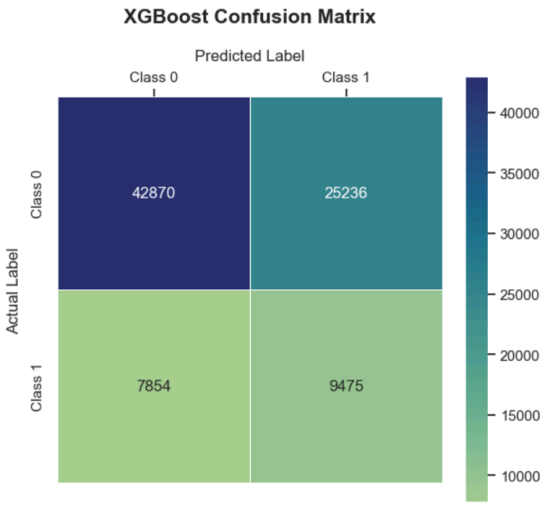
Among the models tested, Random Forest delivered the strongest balance between detection capability and precision, achieving a recall of 0.60 and a PR AUC of 0.32.

It also achieved the highest balanced accuracy, indicating consistent performance across both delayed and on-time flights.

Model	Recall_Pos	PR_AUC	Balanced_Acc
Random Forest	0.60	0.32	0.60
XGBoost	0.55	0.30	0.59
Logistic Regression	0.65	0.29	0.59
Decision Tree	0.54	0.27	0.56

Shown below is how the top 2 models compare:

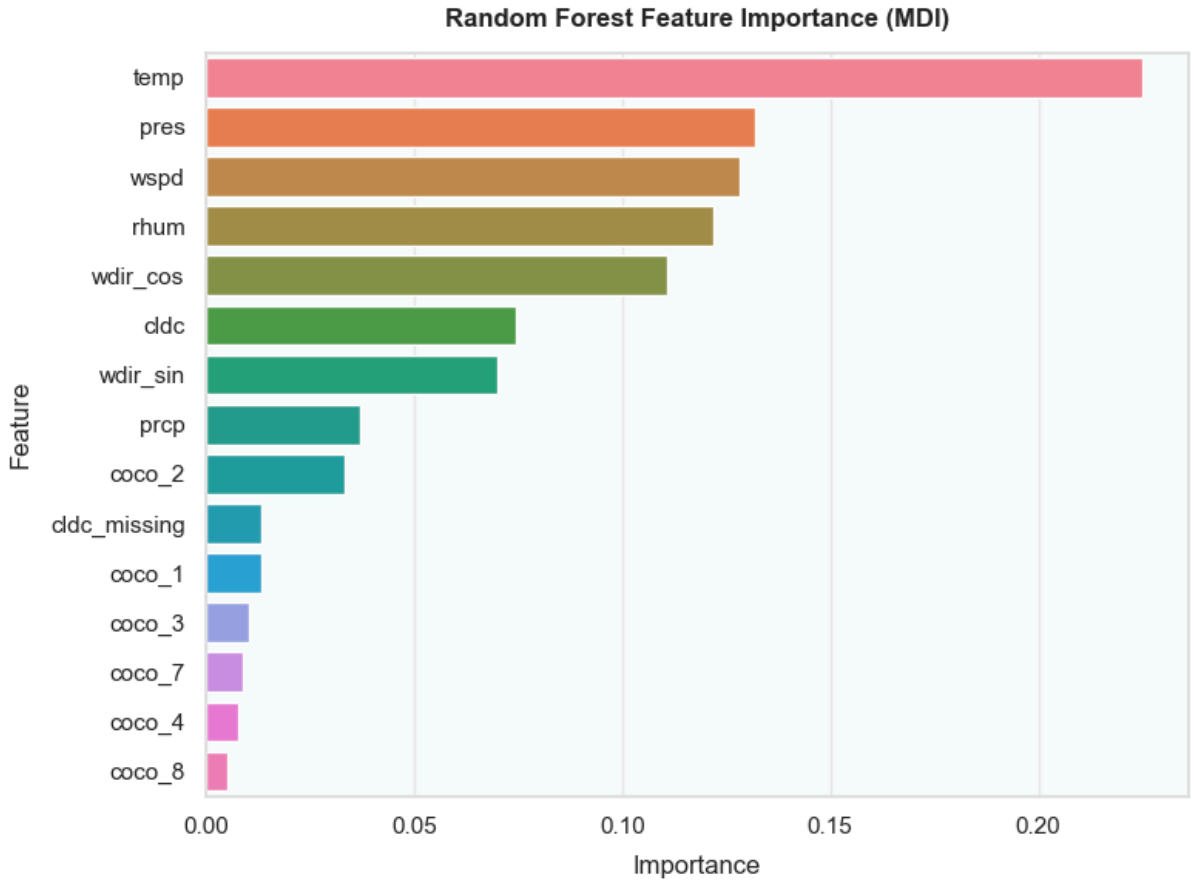




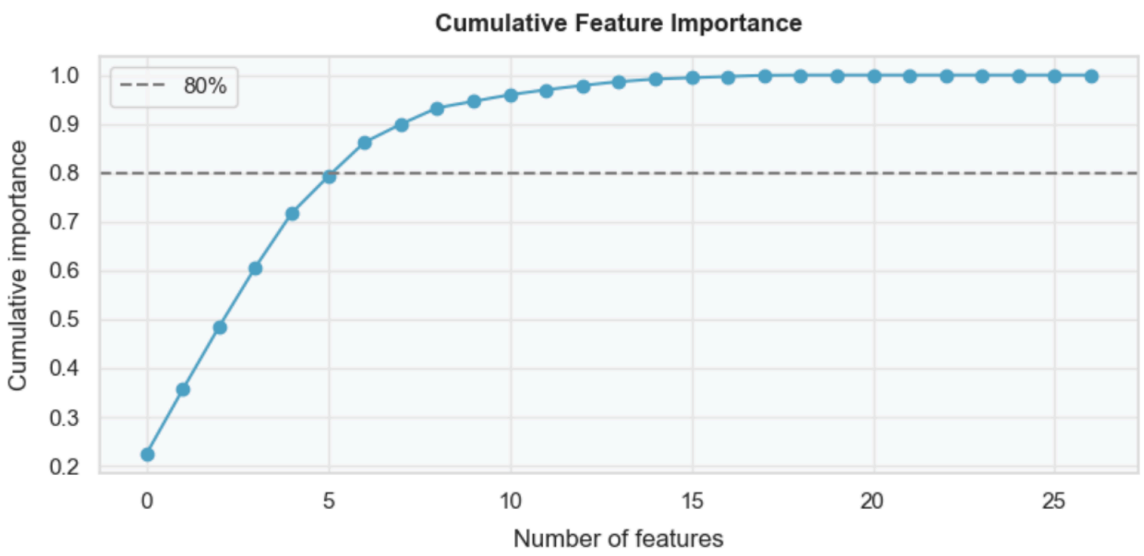
Random Forest improves delay detection by roughly 9% relative to XGBoost, whilst increasing false positives by about 7%. Given that missed delays typically carry greater operational risk than unnecessary alerts, this represents a favourable trade-off.

Feature Importance

The below graph shows Random Forest Feature Importance (MDI).

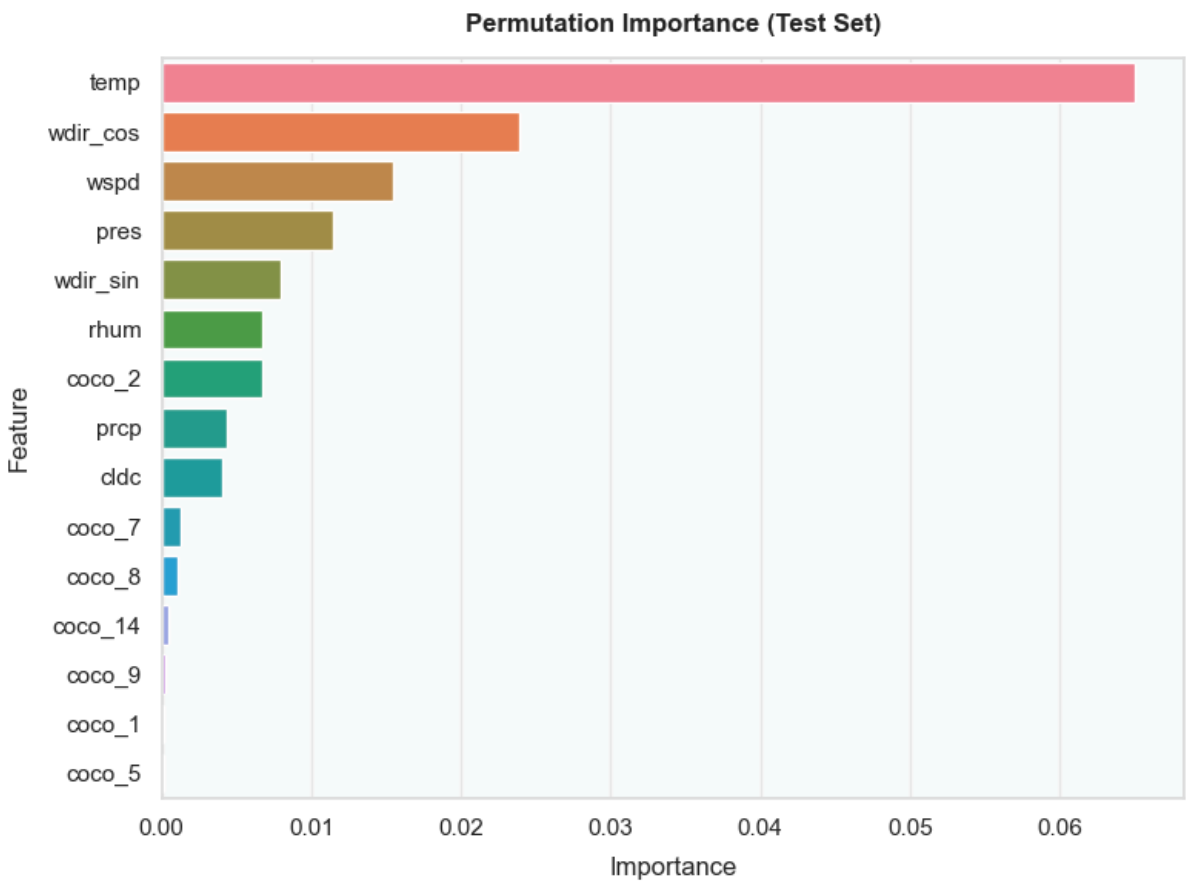


Features that contribute most to reducing impurity in decision trees are temperature, pressure, wind speed, wind direction and humidity are the primary drivers of predicted delay risk. That combination often signals storm systems.



Cumulative Feature Importance graph indicates that approx. 80% of total importance is captured by 5 features. After 10 features, gains are marginal.

The Permutation Importance graph confirms that a small set of continuous meteorological variables accounts for most of the model's predictive power, with temperature contributing the largest marginal effect.



Temperature dominating our model is not random. Temperature often acts as a seasonal proxy, again indicating that the model is sensing weather regimes, not isolated variables.

Outcomes

The objective was to develop a predictive model using combined flight and meteorological data. We explored several machine learning models and the Random Forest model proved to offer the best balance between maximising the number of predicted delays and false positives, and we recommend this model given the industry specifics.

For next steps, we recommend:

- Testing the model on other months, perhaps with a higher ratio of delayed flights
- Testing the model on other origins of departure to see how it holds
- Incorporating other predictors such as days of week, month, carrier, etc.

Implementation

Considerations for implementing the model in production would be making sure that there is reliable data available for both flights and weather in a clean format that replicates the data format the model was trained on.

Data answer

We set out to explore to what extent weather conditions influence the likelihood of flight departure delays, and our dataset included features that were useful in establishing the answer to this problem, which was confirmed by the EDA.

Limiting the dataset to one airport only and adding weather for that location proved to be a good approach since it allowed us to explore the data better and would reflect the real world approach as well.

Business answer

The goal was to catch delayed flights whilst also keeping an eye on the false positives as they also come at cost, so we looked at recall as well as average precision score for metrics.

In the course of this project we achieved our goal of developing a predictive model using combined flight and meteorological data. Our Random Forest model is able to correctly predict 60% of the flight delays based on the weather conditions at the time of scheduled flight departure. Whilst this does not capture every event, many flight disruptions stem from factors beyond weather — such as inbound aircraft delays or carrier-related constraints — which were outside the scope of this model.

Response to stakeholders

The model developed in this project could serve as a helpful tool in detecting upcoming flight departure delays linked to weather conditions for the JFK airport, and it could reduce the pressures and costs associated with flight delays, at the departure airport itself or further down the network.

End-to-end solution

Final output could be a dashboard with upcoming flights information (e.g. 3-7 days out), with meteorological data merged in, and an indicator on which flights are at a risk of being delayed. Even if flights are flagged as at risk of being delayed closer to the departure time, this could allow the stakeholders to make the appropriate operational decisions to minimise the delay impact and costs.

References

Data:

- 3 notebooks with EDAs and model itself:
  - Capstone_EDA_2022-2023.ipynb
  - Capstone_EDA_2025.ipynb
  - Capstone_model_2022_2025.ipynb
- Csv files with the data
- Presentation pdf.

Useful links:

- Data source: https://www.transtats.bts.gov/DL_SelectFields.aspx?gnoyr_VQ=FGJ&QO_fu146_anzr=b0-gvzr
- Meteostat Meteorological Parameters: https://dev.meteostat.net/parameters
- Meteostat Weather Code Conditions: https://dev.meteostat.net/formats.html
- https://www.cirium.com/thoughtcloud/predicting-flight-delays-how-airlines-are-harnessing-ai-to-minimize-disruptions/
- https://travelradar.aero/a-data-driven-analysis-of-the-flight-delays-on-airline-profitability/
- https://ansperformance.eu/economics/cba/standard-inputs/latest/chapters/cost_of_delay.html
- https://weatherspark.com/y/147186/Average-Weather-at-John-F.-Kennedy-International-Airport-New-York-United-States-Year-Round#google_vignette

Available meteorological parameters

ID	Enumeration	Name	Granularity	Unit(s)	Type(s)
`temp`	`Parameter.TEMP`	Air Temperature	**Hourly**	°C	Float64
`rhum`	`Parameter.RHUM`	Relative Humidity	**Hourly**, **Daily**, Monthly, Normals	%	UInt8
`prcp`	`Parameter.PRCP`	Total Precipitation	**Hourly**, **Daily**, **Monthly**, **Normals**	mm	Float64
`snwd`	`Parameter.SNWD`	Snow Depth	**Hourly**, **Daily**	cm	UInt16
`wdir`	`Parameter.WDIR`	Wind Direction	**Hourly**	°	UInt16
`wspd`	`Parameter.WSPD`	Wind Speed	**Hourly**	km/h	Float64
`wpgt`	`Parameter.WPGT`	Peak Wind Gust	**Hourly**, **Daily**	km/h	Float64
`pres`	`Parameter.PRES`	Air Pressure (MSL)	**Hourly**	hPa	Float64
`tsun`	`Parameter.TSUN`	Sunshine Duration	**Hourly**, **Daily**, **Monthly**, **Normals**	min	UInt8, UInt16
`cldc`	`Parameter.CLDC`	Cloud Cover	**Hourly**	okta	UInt8
`coco`	`Parameter.COCO`	Weather Condition Code	**Hourly**	—	UInt8